def paged_attention():

=> paper that introduced vLLM to pagedAttention():

=> ~~Solution~~: idea: Divide the KV cache of a sequence into non-contiguous blocks.

=> sharing the system prompt.

=> sampling multiple responses per prompt

other vLLM optimizations:

①: kernel to fuse block read and attention.

②: use latest kernels (reduce kernel lauch overhead)

(flash attention, flash decoding)

③: use CUDA graphs to avoid kernel lauch overhead

use ideas from operating systems (paging) to make full use of memory for Dynamic workloads.

Final summary:

①: Inference is important (actual use, evaluation, reinforcement learning)

②: Different characteristics compared to training (memory-bound, dynamic)

③: Tenchniques: new architectures, quatization, pruning/distillation, speculative decoding

④: ideas from operating system (vLLM paged Attention, speculative execution)

⑤: New architectures have huge potential for improvement